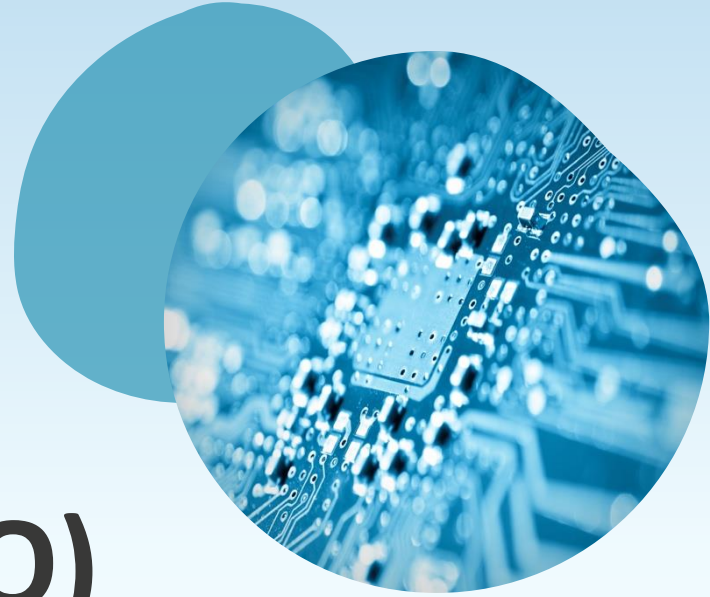


General Purpose Input/Output (GPIO)

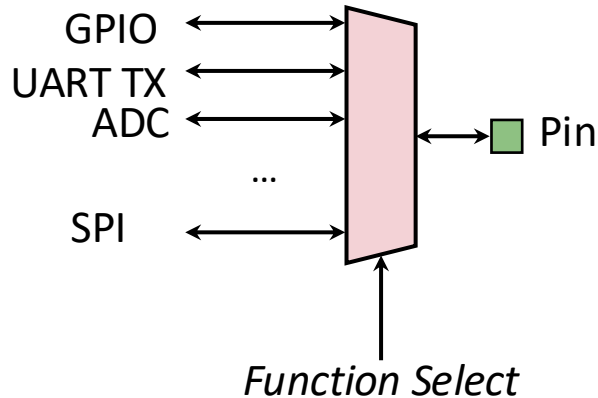
Embedded Software for the Internet Of Things

Kasim Sinan Yildirim



General-purpose and peripheral pin functions

- Pin **multiplexing** selects which internal function reaches the package pin.



GPIO mode

Software reads or drives a digital level.

Typical devices include **LEDs, switches, keypads**, and simple control signals.

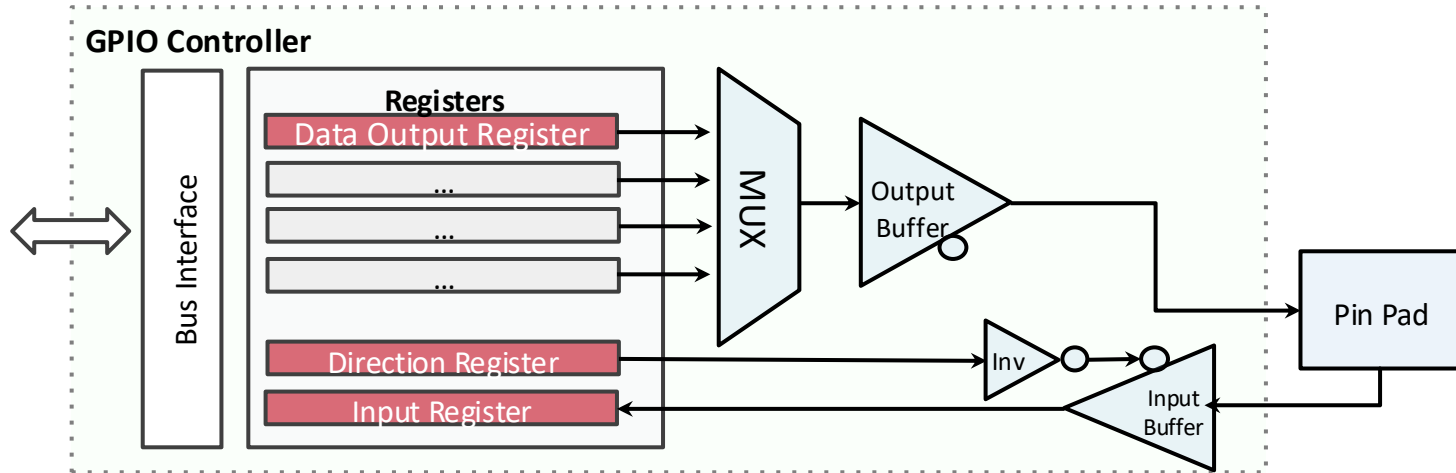
Peripheral mode

The pin connects to on-chip hardware such as a timer, **UART, SPI, I²C**, or **ADC** input.

The peripheral controls the signals and timing.

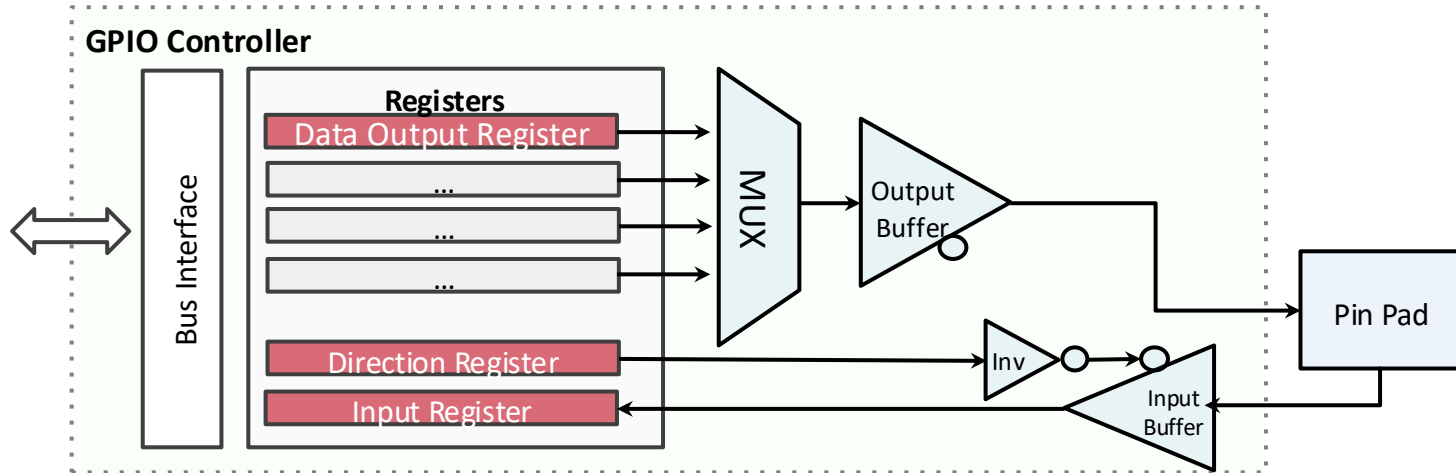
GPIO (General Purpose I/O)

- A **GPIO port** connects memory-mapped registers to configurable electrical circuitry at the package pin.



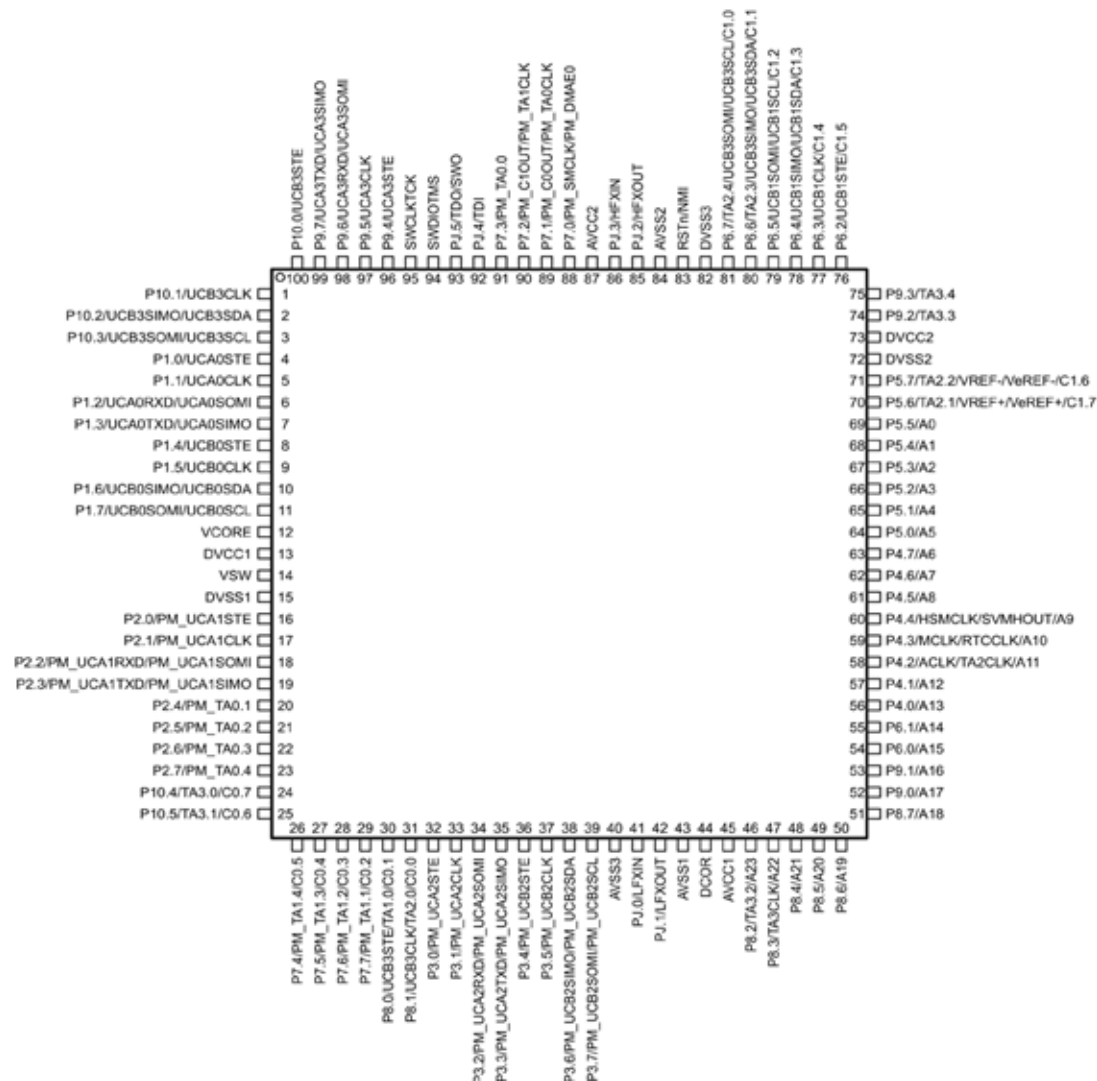
GPIO Direction and Data Registers

- Minimum of 3 registers associated with each of GPIO port:
 - **Data IN**, **Data Out**, and **Direction**
 - The Direction (**PxDIR**) register is used to make the pin either **input** or **output**
 - Data registers to actually write to the pin (**PxOUT**) or read data from the pin (**PxIN**)



MSP GPIO Ports

- The GPIO ports in MSP432 are designated as port **P1** to **P10** and **PJ**
 - P1 to P10 are also referred as the **Simple I/O** or **Digital I/O ports**
 - Port J has special function such as external crystal oscillator and JTAG connections
- Each port contains **eight numbered pins**



Commonly Used Registers of Port1

- The **base address** of I/O port is 0x4000 4C00

Address	Name	Description	Type	Reset Value
0x4000 4C00	P1IN	Port1 Input Register	R	0b00000000
0x4000 4C02	P1OUT	Port1 Output Register	R/W	0b00000000
0x4000 4C04	P1DIR	Port1Direction Register	R/W	0b00000000
0x4000 4C06	P1REN	Port1 Resistor Enable Register	R/W	0b00000000
0x4000 4C08	P1DS	Port1 Drive Strength Register	R/W	0b00000000
0x4000 4C0A	P1SEL0	Port1Select 0 Register	R/W	0b00000000
0x4000 4C0C	P1SEL1	Port1Select 1 Register	R/W	0b00000000

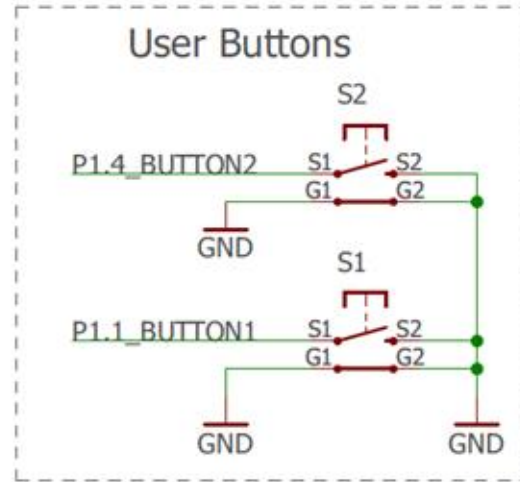
Bit-mask Operations

- Use read-modify-write operations when other pins in the same port must **retain their configuration**.

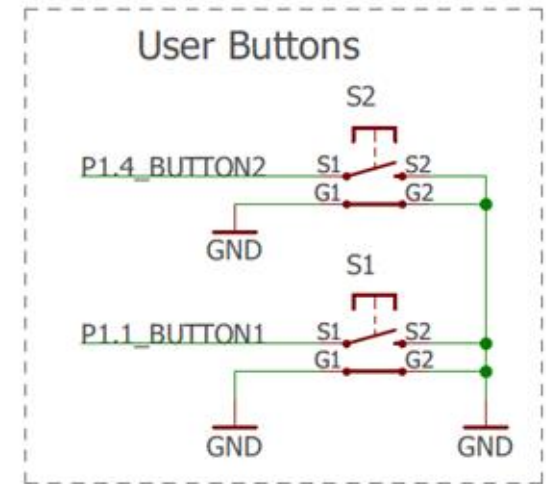
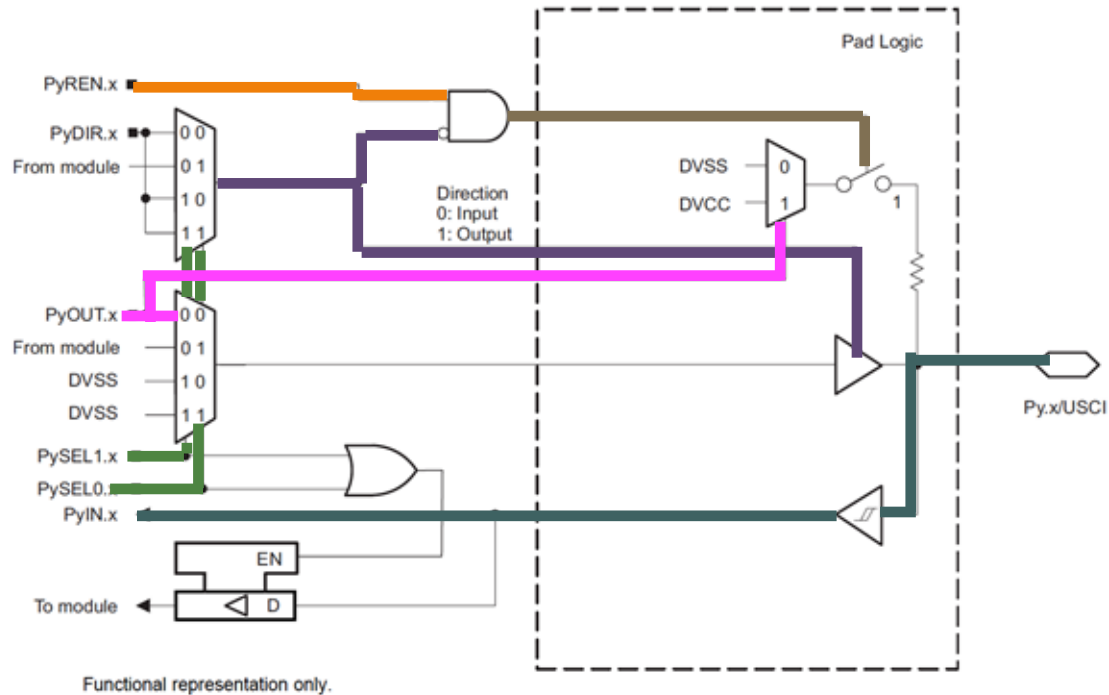
Operation	Expression	Effect
Set selected bits	<code>REG = MASK</code>	Selected bits become 1
Clear selected bits	<code>REG &= ~MASK</code>	Selected bits become 0
Toggle selected bits	<code>REG ^= MASK</code>	Selected bits invert
Read selected bits	<code>REG & MASK</code>	Nonzero when any selected bit is 1
Replace whole register	<code>REG = value</code>	Every bit receives a new value

GPIO as Input

- MSP432 LAUNCHPAD
 - Buttons are connected to P1.1 AND P1.4



Configure User Button (P1.1)



Configure User Button (P1.1)

- Configure P1.1 as input port
 - **BIT1** = 0000 0001

```
/* Configure P1.1 as input*/  
P1->DIR = ~BIT1;
```

10.4.4 PxDIR Register

Port X Direction Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 10-4. PxDIR Register

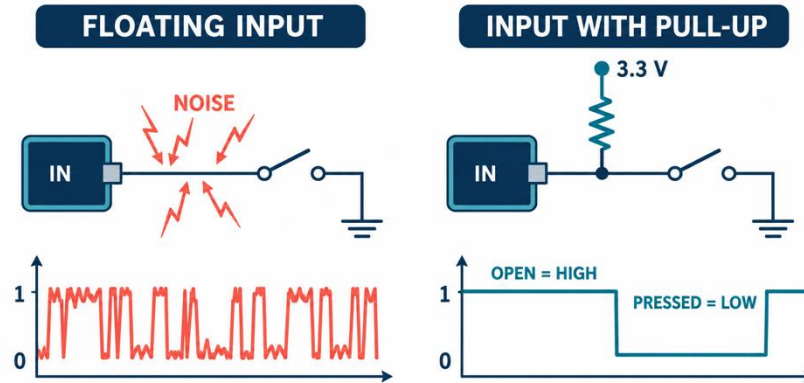
7	6	5	4	3	2	1	0
PxDIR							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 10-7. PxDIR Register Description

Bit	Field	Type	Reset	Description
7-0	PxDIR	RW	0h	Port X direction. 0b = Port configured as input 1b = Port configured as output

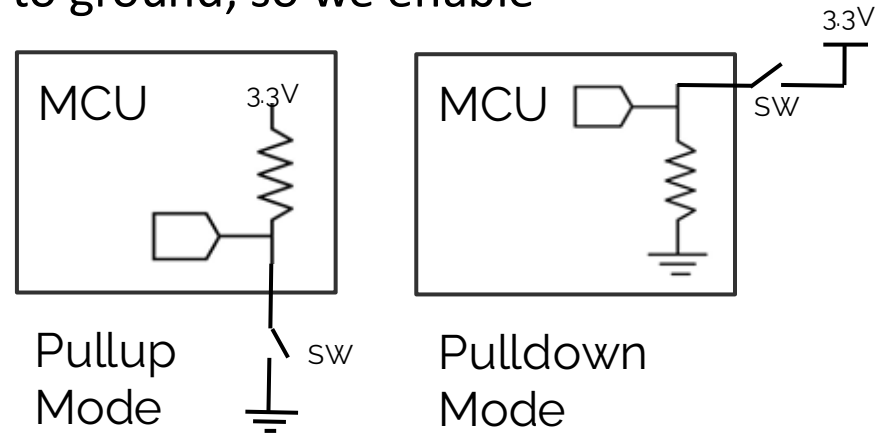
Pullup and Pulldown Resistors

- If we do not use internal **pullup** (or **pulldown**) **resistors**, we have a problem.
 - We need to ensure a known value on the output if a pin is left **floating**



Pullup and Pulldown Resistors

- If we do not use internal **pullup** (or **pulldown**) **resistors**, we have a problem.
 - We need to ensure a known value on the output if a pin is left **floating**
- We want the switch SW to pull the pin to ground, so we enable the pull-up
 - The pin value is:
 - High when SW is not pressed
 - Low when SW is pressed



Configure User Button (P1.1)

- Select Pullup mode.

```
/* Configure P1.1 as input*/  
P1->DIR = ~BIT1;  
/* Select pullup*/  
P1->OUT = BIT1;
```

Figure 10-3. PxOUT Register

7	6	5	4	3	2	1	0
PxOUT							
rw	rw	rw	rw	rw	rw	rw	rw

Table 10-6. PxOUT Register Description

Bit	Field	Type	Reset	Description
7-0	PxOUT	RW	Undefined	Port X output. When I/O configured to output mode: 0b = Output is low. 1b = Output is high. When I/O configured to input mode and pullups/pulldowns enabled: 0b = Pulldown selected 1b = Pullup selected

Configure User Button (P1.1)

- Enable Pullup Resistor

```
/* Configure P1.1 as input*/  
P1->DIR = ~BIT1;  
/* Select pullup*/  
P1->OUT = BIT1;  
/* Enable pullup*/  
P1->REN = BIT1;
```

Figure 10-5. PxREN Register

7	6	5	4	3	2	1	0
PxREN							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 10-8. PxREN Register Description

Bit	Field	Type	Reset	Description
7-0	PxREN	RW	0h	Port X pullup or pulldown resistor enable. When the port is configured as an input, setting this bit enables or disables the pullup or pulldown. 0b = Pullup or pulldown disabled 1b = Pullup or pulldown enabled

Configure User Button (P1.1)

- Set GPIO Mode

Figure 10-7. PxSEL0 Register

7	6	5	4	3	2	1	0
PxSEL0							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 10-10. PxSEL0 Register Description

Bit	Field	Type	Reset	Description
7-0	PxSEL0	RW	0h	Port function selection. Each bit corresponds to one channel on Port X. The values of each bit position in PxSEL1 and PxSEL0 are combined to specify the function. For example, if P1SEL1.5 = 1 and P1SEL0.5 = 0, then the secondary module function is selected for P1.5. See PxSEL1 for the definition of each value.

Figure 10-8. PxSEL1 Register

7	6	5	4	3	2	1	0
PxSEL1							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 10-11. PxSEL1 Register Description

Bit	Field	Type	Reset	Description
7-0	PxSEL1	RW	0h	Port function selection. Each bit corresponds to one channel on Port X. The values of each bit position in PxSEL1 and PxSEL0 are combined to specify the function. For example, if P1SEL1.5 = 1 and P1SEL0.5 = 0, then the secondary module function is selected for P1.5. 00b = General-purpose I/O is selected 01b = Primary module function is selected 10b = Secondary module function is selected 11b = Tertiary module function is selected

```
/* Configure P1.1 as input*/  
P1->DIR = ~BIT1;  
/* Select pullup*/  
P1->OUT = BIT1;  
/* Enable pullup*/  
P1->REN = BIT1;  
/* Set as I/O */  
P1->SEL0 = 0;  
P1->SEL1 = 0;
```

Alternate functions

- Pins may have **alternate functions**:
 - UART, SPI, and I2C, timers...
- SEL1 and SEL0 choose
 - The function **multiplexer** selects GPIO or alternate function

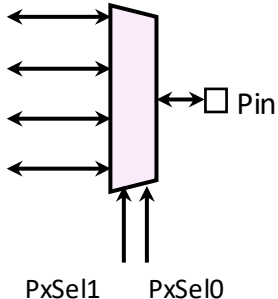
SEL1	SEL0	Selected function
0	0	Simple digital I/O
0	1	Primary module function
1	0	Secondary module function
1	1	Tertiary module function

GPIO

Primary Module Function

Secondary Module Function

Tertiary Module Function



Pin	Other functionality
P1.2 / P1.3	UART receive and transmit, I ² C signals
P2 pins	Timer capture or compare outputs
P4 pins	Clock, timer, and peripheral bus signals
P5 / P6 pins	Analog inputs, references, and communication signals
P7 to P10	Timer, clock, communication, and analog functions

Configure User Button (P1.1)

- Check Input Register to detect the input

Figure 10-2. PxIN Register

7	6	5	4	3	2	1	0
PxIN							
r	r	r	r	r	r	r	r

Table 10-5. PxIN Register Description

Bit	Field	Type	Reset	Description
7-0	PxIN	R	Undefined	Port X Input 0b = Input is low 1b = Input is high

```
/* Configure P1.1 as input*/
P1->DIR = ~BIT1;
/* Select pullup*/
P1->OUT = BIT1;
/* Enable pullup*/
P1->REN = BIT1;
/* Set as I/O */
P1->SEL0 = 0;
P1->SEL1 = 0;

...

/* Catch Button Press */
while (P1->IN & BIT1);
while (!(P1->IN & BIT1));

...
```

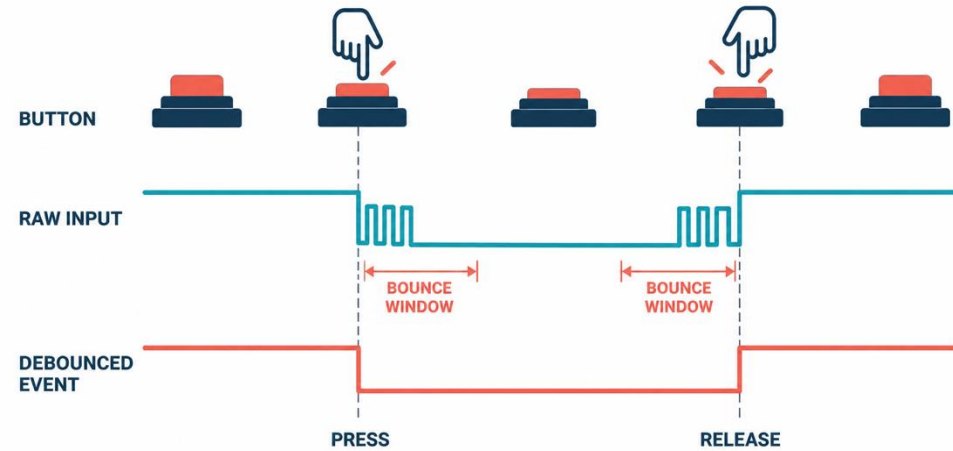
Lab #1

- Put breakpoints and observe the execution.

```
int main(void) {  
    P1->SEL1 &= ~2;           /* configure P1.1 as simple I/O */  
    P1->SEL0 &= ~2;  
    P1->DIR &= ~2;           /* P1.1 set as input */  
    P1->REN |= 2;            /* P1.1 pull resistor enabled */  
    P1->OUT |= 2;            /* Pull up/down is selected by P1->OUT */  
    while (1) {  
        while(P1->IN & 2);  
        while (!(P1->IN & 2));  
    }  
}
```

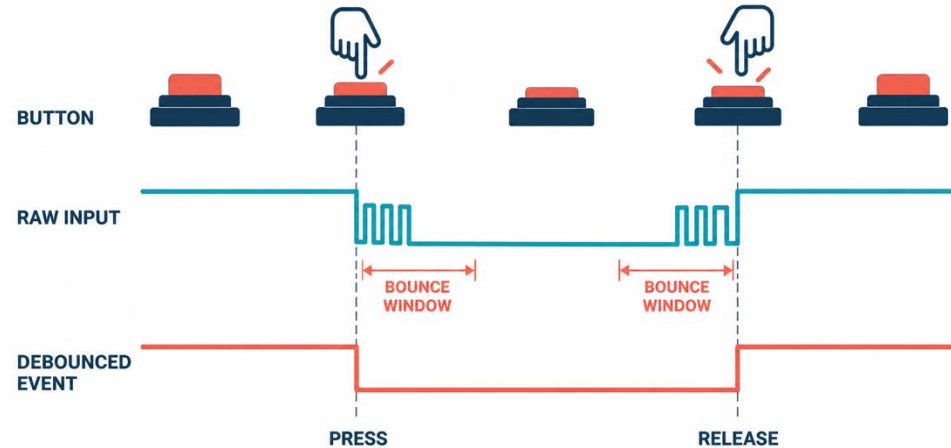
Bouncing GPIO Input

- Mechanical **bounce** creates several digital transitions.



Bouncing GPIO Input

- Mechanical **bounce** creates several digital transitions.

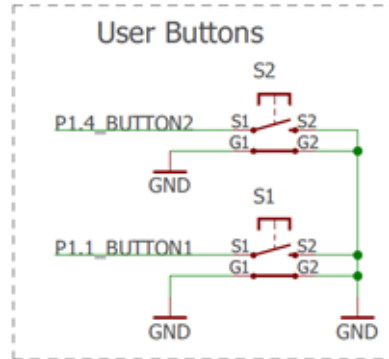


How to handle in software?

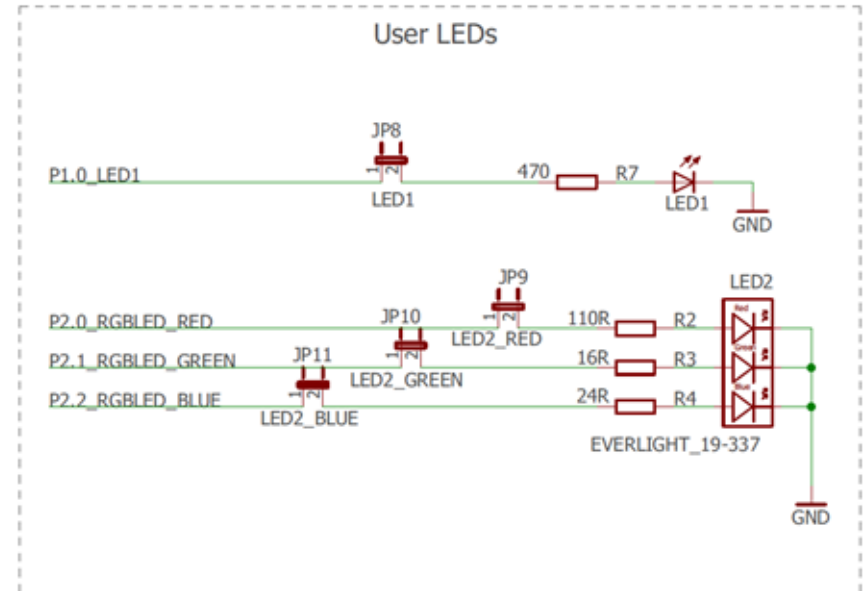
Strategy	How it works
Delay after edge	Wait, then sample again
Stable sample count	Accept change after N identical samples

Lab #2

- Write the code that toggles Green LED when S1 switch is pushed.

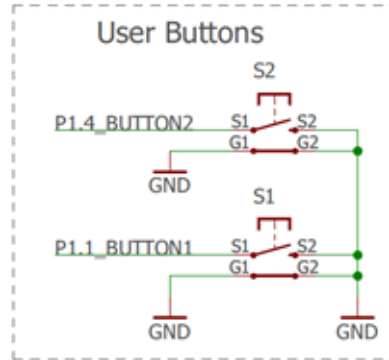


Buttons and LEDs

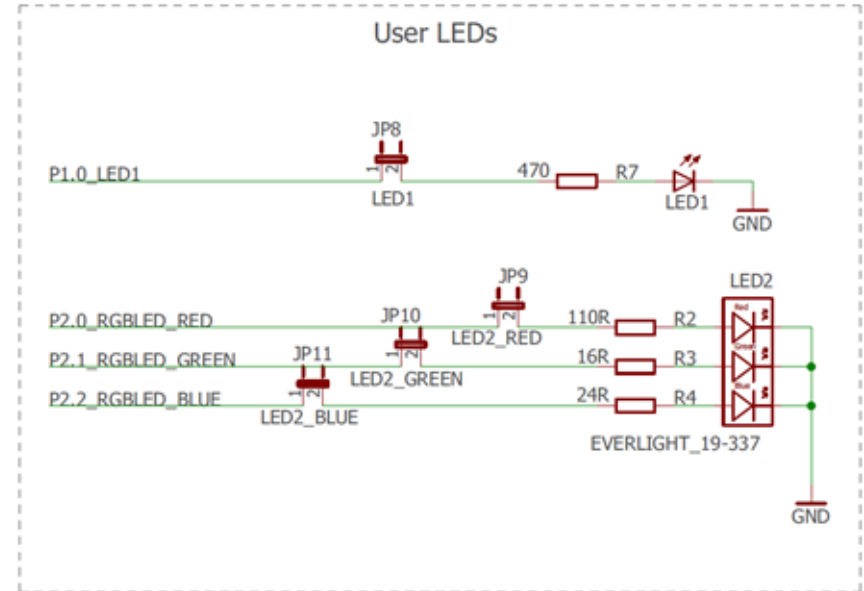


Lab #3

- Write the code that toggles Green LED when S1 switch is pushed and toggles Blue LED when S2 is pushed.



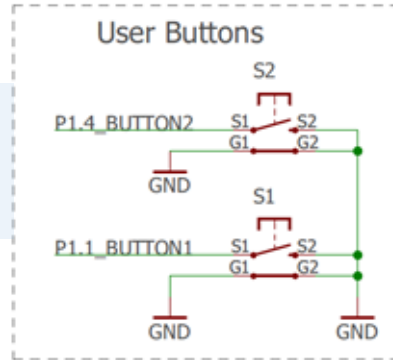
Buttons and LEDs



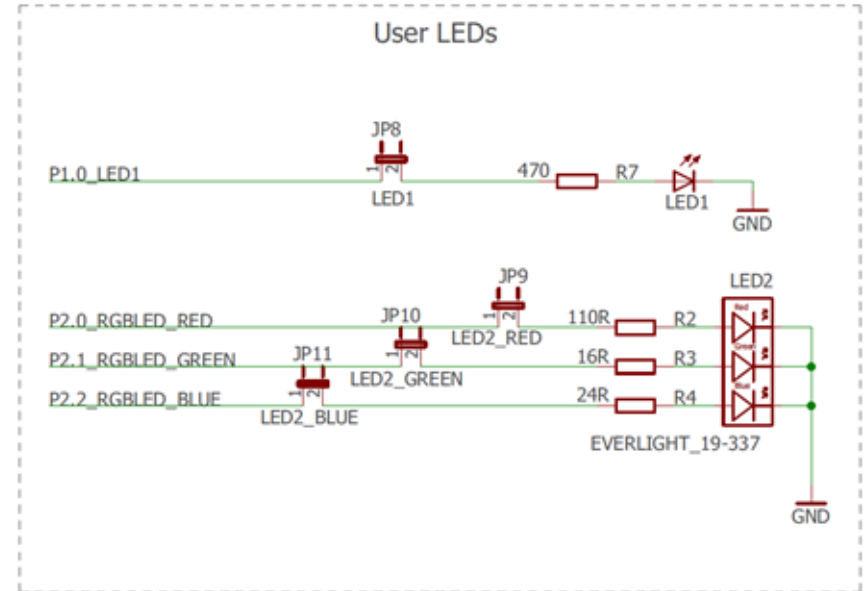
Lab #3

- Write the code that toggles Green LED when S1 switch is pushed and toggles Blue LED when S2 is pushed.

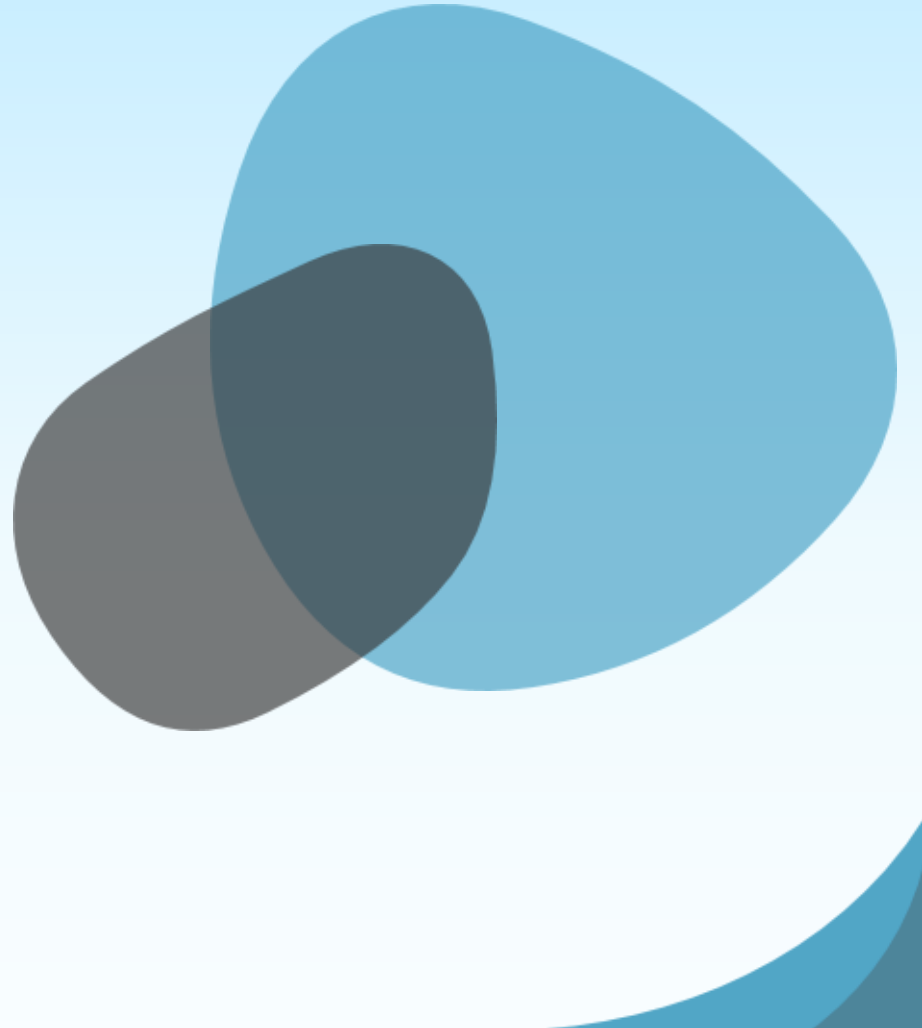
Could you detect switch actions simultaneously?



Buttons and LEDs



MSP432 Port Reference



Commonly Used Registers of Port2

Address	Name	Description	Type	Reset Value
0x4000 4C01	P2IN	Port2 Input Register	R	0b00000000
0x4000 4C03	P2OUT	Port2 Output Register	R/W	0b00000000
0x4000 4C05	P2DIR	Port2 Direction Register	R/W	0b00000000
0x4000 4C07	P2REN	Port2 Resistor Enable Register	R/W	0b00000000
0x4000 4C09	P2DS	Port2 Drive Strength Register	R/W	0b00000000
0x4000 4C0B	P2SEL0	Port2 Select 0 Register	R/W	0b00000000
0x4000 4C0D	P2SEL1	Port2 Select 1 Register	R/W	0b00000000

Commonly Used Registers of Port3

Address	Name	Description	Type	Reset Value
0x4000 4C20	P3IN	Port3 Input Register	R	0b00000000
0x4000 4C22	P3OUT	Port3 Output Register	R/W	0b00000000
0x4000 4C24	P3DIR	Port3 Direction Register	R/W	0b00000000
0x4000 4C26	P3REN	Port3 Resistor Enable Register	R/W	0b00000000
0x4000 4C28	P3DS	Port3 Drive Strength Register	R/W	0b00000000
0x4000 4C2A	P3SEL0	Port3 Select 0 Register	R/W	0b00000000
0x4000 4C2C	P3SEL1	Port3 Select 1 Register	R/W	0b00000000

Commonly Used Registers of Port4

Address	Name	Description	Type	Reset Value
0x4000 4C21	P4IN	Port4 Input Register	R	0b00000000
0x4000 4C23	P4OUT	Port4 Output Register	R/W	0b00000000
0x4000 4C25	P4DIR	Port4 Direction Register	R/W	0b00000000
0x4000 4C27	P4REN	Port4 Resistor Enable Register	R/W	0b00000000
0x4000 4C29	P4DS	Port4 Drive Strength Register	R/W	0b00000000
0x4000 4C2B	P4SEL0	Port4 Select 0 Register	R/W	0b00000000
0x4000 4C2D	P4SEL1	Port4 Select 1 Register	R/W	0b00000000

Alternate Pin Functions

PxSEL1.y	PxSEL0.y	Meaning
0	0	Alternative 0 (Default) Simple I/O
0	1	Alternative 1 (UART, SPI, I2C, ...)
1	0	Alternative 2 (Timers, ...)
1	1	Alternative 3 (ADC, Comparator, ...)

Port 1 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P1.0	Simple I/O	UCA0STE	-	-
P1.1	Simple I/O	UCA0CLK	-	-
P1.2	Simple I/O	UCA0RXD/UCA0SOMI	-	-
P1.3	Simple I/O	UCA0TXD/UCA0SIMO	-	-
P1.4	Simple I/O	UCB0STE	-	-
P1.5	Simple I/O	UCB0CLK	-	-
P1.6	Simple I/O	UCB0SIMO/UCB0SDA	-	-
P1.7	Simple I/O	UCB0SOMI/UCB0SCL	-	-

Port 2 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P2.0	Simple I/O	PM_UCA1STE	-	-
P2.1	Simple I/O	PM_UCA1CLK	-	-
P2.2	Simple I/O	PM_UCA1RXD/ PM_UCA1SOMI	-	-
P2.3	Simple I/O	PM_UCA1TXD/ PM_UCA1SIMO	-	-
P2.4	Simple I/O	PM_TA0.1	-	-
P2.5	Simple I/O	PM_TA0.2	-	-
P2.6	Simple I/O	PM_TA0.3	-	-
P2.7	Simple I/O	PM_TA0.4	-	-

Port 3 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P3.0	Simple I/O	PM_UCA2STE	-	-
P3.1	Simple I/O	PM_UCA2CLK	-	-
P3.2	Simple I/O	PM_UCA2RXD/ PM_UCA2SOMI	-	-
P3.3	Simple I/O	PM_UCA2TXD/ PM_UCA2SIMO	-	-
P3.4	Simple I/O	PM_UCB2STE	-	-
P3.5	Simple I/O	PM_UCB2CLK	-	-
P3.6	Simple I/O	PM_UCB2SIMO/ PM_UCB2SDA	-	-
P3.7	Simple I/O	PM_UCB2SOMI/ PM_UCB2SCL	-	-

Port 4 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P4.0	Simple I/O	-	-	A13
P4.1	Simple I/O	-	-	A12
P4.2	Simple I/O	ACLK	TA2CLK	A11
P4.3	Simple I/O	MCLK	RTCCLK	A10
P4.4	Simple I/O	HSMCLK	SVMHOUT	A9
P4.5	Simple I/O	-	-	A8
P4.6	Simple I/O	-	-	A7
P4.7	Simple I/O	-	-	A6

Port 5 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P5.0	Simple I/O	-	-	A5
P5.1	Simple I/O	-	-	A4
P5.2	Simple I/O	-	-	A3
P5.3	Simple I/O	-	-	A2
P5.4	Simple I/O	-	-	A1
P5.5	Simple I/O	-	-	A0
P5.6	Simple I/O	TA2.1	-	VREF+/VeREF+/C1.7
P5.7	Simple I/O	TA2.2	-	VREF-/VeREF-/C1.6

Port 6 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P6.0	Simple I/O	-	-	A15
P6.1	Simple I/O	-	-	A14
P6.2	Simple I/O	UCB1STE	-	C1.5
P6.3	Simple I/O	UCB1CLK	-	C1.4
P6.4	Simple I/O	UCB1SIMO/ UCB1SDA	-	C1.3
P6.5	Simple I/O	UCB1SOMI/ UCB1SCL	-	C1.2
P6.6	Simple I/O	TA2.3	UCB3SIMO/ UCB3SDA	C1.1
P6.7	Simple I/O	TA2.4	UCB3SOMI/ UCB3SCL	C1.0

Port 7 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P7.0	Simple I/O	PM_SMCLK/ PM_DMAE0	-	-
P7.1	Simple I/O	PM_C0OUT/ PM_TA0CLK	-	-
P7.2	Simple I/O	PM_C1OUT/ PM_TA1CLK	-	-
P7.3	Simple I/O	PM_TA0.0	-	-
P7.4	Simple I/O	PM_TA1.4	-	C0.5
P7.5	Simple I/O	PM_TA1.3	-	C0.4
P7.6	Simple I/O	PM_TA1.2	-	C0.3
P7.7	Simple I/O	PM_TA1.1	-	C0.2

Port 8 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P8.0		UCB3STE	TA1.0	C0.1
P8.1		UCB3CLK	TA2.0	C0.0
P8.2		TA3.2		C0.0
P8.3		TA3CLK		A22
P8.4				A21
P8.5				A20
P8.6				A19
P8.7				A18

Port 9 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P9.0				A17
P9.1				A16
P9.2		TA3.3		
P9.3		TA3.4		
P9.4		UCA3STE		
P9.5		UCA3CLK		
P9.6		UCA3RXD/ UCA3SOMI		
P9.7		UCA3TXD/ UCA3SIMO		

Port 10 Alternative Pin Functions

Pin Name	SEL = 00	SEL = 01	SEL = 10	SEL = 11
P10.0		UCB3STE		
P10.1		UCB3CLK		
P10.2		UCB3SIMO/ UCB3SDA		
P10.3		UCB3SOMI/ UCB3SCL		
P10.4		TA3.0		C0.7
P10.5		TA3.1		C0.6